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PERFORMANCE SPECIFICATION

SPLICE, FIBER OPTIC CABLE, GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers environmental- and non-environmental-resistant (chemical, mechanical, and fusion) splices appropriate for military use with optical cables and fibers. Fiber and cable splices specified herein cover a family of general-purpose interconnection hardware providing a variety of compatible optical arrangements.

1.2 Part or identifying number (PIN). PINs to be used for splices acquired to this specification are created as follows:

M	**24623**	**/**	**XX**	**-**	**XXXX**
Prefix for Military Specification	Specification Number	/	Specification Sheet Number	-	One- to Four-Digit Code (see 3.1)

Example: M24623/04-01, M24623/06-1, or M24623/07-2050.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

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AMSC N/A FSC 6060

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1373 - Screw-Thread, Modified, 60 Degrees Stub, Double

MIL-STD-1678-2 - Fiber Optic Cabling Systems Requirements and Measurements (Part 2: Optical Measurements) (Part 2 of 6 Parts)

MIL-STD-1678-3 - Fiber Optic Cabling Systems Requirements and Measurements Physical, Mechanical, Environmental and Material Measurements (Part 3 of 6 Parts)

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-454 - General Guidelines for Electronic Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E595 - Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment

(Copies of this document are available online at www.astm.org.)

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

EIA-364-81 - Combustion Characteristics Test Procedure for Electrical Connector Housings, Connector Assemblies and Sockets

TIA-455 - General Requirements for Standard Test Procedures for Optical Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components

TIA/EIA-455-1 - Cable Flexing for Fiber Optic Interconnecting Devices

TIA-455-2 - Impact Test Measurements for Fiber Optic Devices

TIA-455-6 - Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices

TIA-455-13 - Visual and Mechanical Inspection of Fiber Optic Components, Devices, and Assemblies

TIA-455-22 - Ambient Light Susceptibility of Fiber Optic Components

TIA-455-26 - Crush Resistance of Fiber Optic Interconnecting Devices

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TIA-455-35	-	Fiber Optic Component Dust (Fine Sand) Test
TIA-455-36	-	Twist Test for Fiber Optic Connecting Devices
TIA-455-64	-	Procedure for Measuring Radiation-Induced Attenuation in Optical Fibers and Cables

(Copies of these documents are available online at www.tiaonline.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Fiber optic splices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.6 and 6.3). The provisions of 4.5.4 for retention of qualification are included in this requirement.

3.3 Materials. The splice shall be constructed of materials as specified herein and in the specification sheet (see 3.1). In all cases, materials selected for splice usage shall meet all qualification requirements as specified and be of a type and quality to ensure physical, chemical, and optical compatibility with the requirements of this specification. All materials used shall be nontoxic (see 3.3.1), non-nutrient to fungus (see 3.3.9), and manufactured to good workmanship quality (see 3.8). For space flight applications, materials shall not release in excess of 1 percent total mass loss or 0.1 percent collected volatile condensable materials when tested in accordance with ASTM E595.

3.3.1 Toxicity. Materials used shall not give off toxic or explosive fumes when exposed to flame. When evaluated in accordance with 4.10, materials used shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.10 and 6.5).

3.3.2 Prohibited materials. The materials used in the interconnection shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.3.3 Interior parts. The materials used for splice interior parts shall provide 20 years of service. No incompatibility shall exist between the materials employed such that degradation of these materials can result from in-service use or from test exposures as specified herein.

3.3.4 Exterior parts. Exterior parts of the assembled splice, if metallic, shall have a passivated finish that permits the attainment of a suitable surface finish condition in accordance with high workmanship standards and shall be compatible with external coatings or plating of the type and color specified in the specification sheet (see 3.1).

3.3.5 Finish. The resultant finish on all cable splice closures (see 6.4.3) shall meet the requirements herein.

3.3.6 Solvents, adhesives, and cleaning agents. If epoxy or comparable cement is used in the splicing processes, no incompatibility shall exist between the materials employed such that degradation of the materials can result from in-service use or when tested in accordance with the requirements of the life aging test specified in 3.7.2.

3.3.7 Nonmetallic materials. Nonmetallic materials used in the construction of splices shall not be affected by the use of solvents, adhesives, or cleaning agents nor be degraded at the specified environmental conditions.

3.3.8 Liquid materials. Liquid materials shall not be used.

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3.3.9 Fungus resistance. When tested in accordance with 4.7.2.14, polymeric splice materials shall show sparse or very restricted microbial growth and reproduction with minor or inhibited substrate utilization. There shall be little or no chemical, physical, or structural change detectable.

3.3.10 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life-cycle costs.

3.4 Design and construction. The splices shall be of the construction specified (see 3.1 and 6.1).

3.4.1 General. Splice designs may be single fiber or multiple fibers.

3.4.1.1 Fiber splice. The fiber splice (see 6.4.6) shall include the waveguide splice and the fiber splice housing (see 6.1).

3.4.1.1.1 Waveguide splice. The waveguide splice (see 6.4.16) shall optically align the core and cladding of the optical fiber (see 3.6.1 and 3.6.3).

3.4.1.1.2 Fiber splice housing. The fiber splice housing (see 6.4.7) shall restore the environmental and mechanical integrity of the coating or buffer of the optical fiber.

3.4.1.2 Cable splice. The cable splice (see 6.4.2) shall include the cable splice closure and the fiber splices. The cable splice shall seal the cable to meet the environmental requirements specified herein.

3.4.1.2.1 Cable splice closure. The cable splice closure shall include cable seals necessary to restore mechanical strength and environmental protection to the cable. The cable splice closure shall seal the fiber splice against moisture and contamination as specified herein.

3.4.2 Maintainability. The splices shall require no preventative maintenance.

3.4.3 Interchangeability. All splice parts (see 6.4.14) that have the same military PIN shall be physically and functionally interchangeable without need for modification of such items or of the splicing equipment.

3.4.4 Metals. Metals shall be corrosion resistant. Unless otherwise specified (see 3.1), metals shall be nonmagnetic and all exposed corrosion-resistant steel parts shall have a passivated surface that is compatible with external coatings or plating, as specified (see 3.1).

3.4.5 Dissimilar metals. The use of dissimilar metals in intimate contact should be avoided. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided.

3.4.6 Seals. Cable splice closure seals shall provide environmental isolation for splice interior parts including fiber splices. Cable splice closures shall seal to the attached cables. Grommets, O-rings, boots, gaskets, or other sealing methods may be used and shall accomplish their intended purposes and meet all of the requirements specified herein.

3.4.7 Strain relief. The splice enclosures (see 6.4.11) shall accept and retain a cable support or cable strain relief to maintain cable strength continuity from one cable to the other cable (see 3.5.13).

3.4.8 Tools. Tools used to splice fibers and cables shall be as specified in the specification sheet (see 3.1). When specified (see 6.2), the splice manufacturer shall provide the tools.

3.5 Visual and mechanical.

3.5.1 Size. When examined in accordance with 4.7.2.1, the dimensions and dimensional tolerances for the splice parts shall be as specified in the specification sheet (see 3.1).

3.5.2 Weight. When tested in accordance with 4.7.2.2, the weight and weight tolerances of the splice parts shall be as specified in the specification sheet (see 3.1).

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3.5.3 Color. When visually inspected in accordance with 4.7.2.3, the color of the splice parts shall be as specified in the specification sheet (see 3.1).

3.5.4 Identification marking. Unless otherwise specified (see 3.1), all splices or splice packages shall be identified with markings that are permanent, clearly visible, and legible. Markings shall be legible to the extent that none are missing (in whole or in part), faded, blurred, smeared, or shifted (dislodged), and shall be readily readable. There shall be contrast between characters and the surface. Markings shall be permanent to the extent of withstanding cleaning procedures and of withstanding environmental and mechanical performance tests conducted. Identification marking information shall include the PIN and either the manufacturer's CAGE code, name, or logo (see 4.7.2.4).

3.5.4.1 JAN and J marking. The United States Government has adopted and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to and meet all the requirements of specifications. Accordingly, items acquired to, and meeting all of the criteria specified herein and in applicable specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets, the manufacturer shall remove completely the military part number and the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark "JAN" and Registration Number 2,577,735 for the certification mark "J".

3.5.5 Cable seal flexing (cable splice only). When tested in accordance with 4.7.2.5, splice strain relief mechanisms shall prevent loss of environmental sealing or other damage that may impair the splice operation.

3.5.6 Twist (cable splice only). When tested in accordance with 4.7.2.6, a visual examination of the splice enclosure shall reveal neither seal impairment nor any other splice damage. The splice shall meet the requirements of 3.6.2 during and after the test.

3.5.7 Axial compressive loading (cable splice only). When tested in accordance with 4.7.2.7, a post-test visual examination of the splice shall reveal no indication of cracking, splitting, tearing, or buckling of the splice.

3.5.8 Crush (cable splice only). When tested in accordance with 4.7.2.8, a visual examination of the splice enclosure shall reveal no cracking, splitting, or other defect that permits environmental penetration. Splice enclosure deformation shall not be considered as splice failure. The splice shall meet the requirements of 3.6.2 during and after the test.

3.5.9 Impact. When tested in accordance with 4.7.2.9, the impact resistive property of the splice shall be determined by post-test visual examination of the splice enclosure. This examination shall reveal no cracking, splitting, or other defect that permits environmental penetration. Splice enclosure deformation shall not be considered as splice failure. The splice shall meet the requirements of 3.6.2 after the impact test.

3.5.10 Vibration. When tested in accordance with 4.7.2.10, the splices shall exhibit no visual evidence of loosening of parts, relative motion between splice and cable parts, nor any other damage that can produce physical distortion or wear and may result in fatigue of the mechanical parts or failure of the splice operation. The splice shall meet the requirements of 3.6.4 during the test.

3.5.11 Mechanical shock. When tested in accordance with 4.7.2.11, the splice shall not be damaged and there shall be no loosening of parts. The splice shall meet the requirements of 3.6.4 during the test and 3.6.2 after the test.

3.5.12 Fiber pull out force. When tested in accordance with 4.7.2.12, the minimum fiber-to-fiber splice pull out strength shall be 14.0 Newton (N) (3.1 pounds [lbs]). The splice shall meet the optical requirements of 3.6.2 during and after the test.

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3.5.13 Cable pull out force. The minimum cable-to-cable splice closure pull out strength shall be 720 N (162 lbs) for splices attaching to the strength members of the cable or 67 N (15 lbs) for splices attaching directly to fiber cable components (see 4.7.2.13). The splice shall show no evidence of cable jacket damage, cable clamp failure, cable-to-splice seal damage, distortion from bending of splice parts, or cable disengagement from the clamp. The splice shall meet the requirements of 3.6.2 during and after the test.

3.6 Optical requirements. The splice shall meet the optical requirements as specified herein. Unless otherwise specified (see 3.1), the center wavelength for all measurements shall be 1,300±20 nanometers (nm).

3.6.1 Insertion loss. The initial insertion loss (see 6.4.8) for tunable splices shall not exceed 0.6 decibel (dB) (untuned) and 0.2 dB (tuned). The initial insertion loss for fixed splices shall be 0.6 dB. Unless otherwise specified (see 3.1), the insertion loss shall not exceed 0.9 dB at any time during testing of the splice (see 4.7.3.1).

3.6.2 Change in optical transmittance. When measured in accordance with 4.7.3.2, the change in optical transmittance during or after any specified environmental or mechanical requirement shall be not greater than 0.30 dB.

3.6.3 Return loss. When measured in accordance with 4.7.3.3, return loss (see 6.4.9) shall be not less than 30 dB.

3.6.4 Optical discontinuity. When measured in accordance with 4.7.3.4, no discontinuity (see 6.4.5) shall occur. A discontinuity is considered to be a reduction of signal strength (optical transmittance) of 0.30 dB or more for a duration of 50 microseconds or more.

3.6.5 Crosstalk. When tested in accordance with 4.7.3.5, the signal power levels or the sum of the power levels in the passive output channel or channels shall be below the output signal level of the active channel by not less than 50 dB.

3.6.6 Ambient light susceptibility. When tested in accordance with 4.7.3.6, the optical power of the light from the fibers (after accounting for cable losses and optical junction losses between the fiber and the detector) shall be less than -50 decibel-milliwatt (dBm) (dB reference to 1 milliwatt [mW]).

3.7 Environmental requirements.

3.7.1 Environmental properties. The splice shall meet all the specified requirements (see 3.1) during the specified operating environments and after the specified storage environments. The operating and storage temperature ranges shall be as specified (see 3.1) in table I.

TABLE I. Temperature ranges.

Range	Operating temperature (°C)	Non-operating temperature (°C)	Storage temperature (°C)
1	-28 to +65	-40 to +70	-40 to +70
2	-55 to +165	-40 to +85	-40 to +85

3.7.2 Life aging. When tested in accordance with 4.7.4.1, the splices subjected to these specified accelerated aging exposures shall not exhibit visual evidence of dimensional change, opening of seals, cracking or crazing of components or finishes, identification marking impairment, leakage of waterproofing compounds, or other defects detrimental to the operation of the splice. The splice shall meet the requirements of 3.6.2 after the test.

3.7.3 Thermal shock. When tested in accordance with 4.7.4.2, a post-test visual examination of the test splices shall reveal no leakage of waterproofing compounds or other apparent loss of sealing capability, no surface or identification marking impairment, or any damage detrimental to the operation of the splice. The splice shall meet the requirements of 3.6.2 after the test.

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3.7.4 Temperature/humidity cycling. When subjected to high humidity and cyclic temperature exposure in accordance with 4.7.4.3, the splice parts shall not swell, have impaired identification markings, or degrade such that splice performance is impaired. The splice shall meet the requirements of 3.6.2 during and after the test.

3.7.5 Salt spray (corrosion) (cable splice only). When tested in accordance with 4.7.4.4, no visible evidence of salt penetration in splice enclosures shall be observed and no corrosive effects shall be seen on the external splice parts.

3.7.6 Water pressure (cable splice only). When tested in accordance with 4.7.4.5, visual inspection of the test splice shall reveal no penetration of water in the sealed region of the splice. The splice shall meet the optical requirements of 3.6.2 after the test.

3.7.7 Freezing water immersion (cable splice only). When tested in accordance with 4.7.4.6, the splice shall not be damaged during exposure. The splice shall meet the optical requirements of 3.6.2 during and after the test.

3.7.8 Sand and dust (cable splice only). When tested in accordance with 4.7.4.7, the splice shall not be damaged during exposure. The splice shall meet the optical requirements of 3.6.2 during and after the test.

3.7.9 Nuclear radiation resistance (lensed splice only). When tested in accordance with 4.7.4.8, the splices shall not be damaged during exposure. The splice shall meet the requirements of 3.6.2 during and after the test, where the maximum change in transmittance shall be 5.0 dB and the change in transmittance at the specified recovery time of 1,000 seconds shall be as follows:

a. Not greater than 3.0 dB at -28 °C.

b. Not greater than 2.0 dB at 28 °C.

c. Not greater than 2.0 dB at 85 °C.

3.7.10 Fluid immersion (cable splice only). When tested in accordance with 4.7.4.9, visual examination of the splice shall reveal no swelling or softening of material, no loss of sealing capability or identification marking, and no discoloration or effects detrimental to the intended use of the splice.

3.7.11 Flammability. When tested in accordance with 4.7.4.10, burning and after-glow of the splice shall cease within 30 seconds after removal of the applied flame, and there shall be no violent burning or explosive-type fire. There shall be no dripping that will cause the flammable material to ignite. The splice shall meet the optical requirements of 3.6.2 during and after the first application of the flame.

3.7.12 Temperature cycling. When tested in accordance with 4.7.4.11, the splice shall not exhibit visual evidence of loss of sealing capability and shall meet the requirements of 3.6.2 during and after the test.

3.8 Workmanship. All details of workmanship shall be as specified herein, when examined in accordance with 4.8. The splices shall be dimensionally uniform and free of manufacturing flaws that would degrade performance, would inhibit proper assembly or mating, or would otherwise yield an inferior product. Minimum workmanship requirements are listed below and shall be verified by visual examination (see 4.8). This list is not intended to restrict other pertinent workmanship requirements deemed necessary by the contractor.

a. Splice parts that adversely affect environmental sealing or degrade the optical fiber alignment shall not be permitted.

b. Cuts, abrasions, holes, thin spots, peeling or chipping of plating or finish, nicks, burns, or other substandard surface blemishes shall not be permitted.

3.9 Screw threads. When tested in accordance with 4.9, coupling screw threads shall be in accordance with MIL-STD-1373 and as specified (see 3.1). Slight out-of-roundness beyond the tolerances specified is acceptable if the threads can be checked without forcing the thread gauges. Screw threads may be relieved, provided the relief does not interfere with proper performance of the screw threads.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.5).

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b. Conformance inspection (see 4.6).

4.2 Verification program. Requirements for the verification program shall be as identified in the qualification instructions (see 6.3.2).

4.3 Test equipment and inspection facilities. Requirements for test equipment and inspection facilities shall be as identified in the qualification instructions (see 6.3.3).

4.4 Inspection conditions. Unless otherwise specified (see 3.1), all inspections shall be performed in accordance with the standard test conditions specified in TIA-455 or as specified herein.

4.5 Qualification inspection. Qualification inspection shall include the examinations of 4.7.2 through 4.7.2.4, 4.8, and 4.9 and the tests of 4.7.2.5 through 4.7.4.11, as applicable.

4.5.1 Sample. Fiber optic splice samples complying with the requirements specified herein and in the specification sheets (see 3.1) shall be submitted for qualification testing. The sample shall consist of two sample units for group II and two sample units for groups III and IV (lensed splices only) as specified in table II.

4.5.1.1 Sample unit. A sample unit shall consist of the parts required to make a fully assembled fiber optic splice submitted for qualification.

4.5.1.2 Sample unit preparation. Unless otherwise specified herein and except for group I inspection of table II and, splices shall be fully assembled utilizing two lengths of cable, each at least 5 meters long, compatible with the splice undergoing the test. The cable shall contain at least two fibers for units that have the capability to splice more than one fiber. At least two adjacent splices shall be made for each multisplice unit. When optical crosstalk tests are required, crosstalk shall be measured between fibers/splices enclosed in the same sample unit.

4.5.2 Inspection routine. The samples shall be subjected to the qualification inspection specified in table II. Group I inspections may be performed in any order; however, group II and III inspections shall be performed in the order shown. In cases where certain requirements and tests are applicable only when specified (see 3.1), these tests shall be conducted in the order shown when specified in the appropriate specification sheet (see 3.1). Tests that are specified as not applicable to a specific splice construction shall not be conducted. All sample units shall be subjected to the group I inspections. Test sample units shall then be subjected to the group II and III tests specified in table II; however, each test specimen shall be subjected to only one group of tests in addition to the tests of group I.

4.5.3 Failures. Failure of any sample or specimen to meet the requirements specified herein shall be cause for disapproval of or removal of product qualification.

4.5.4 Retention of qualification. To retain qualification, the manufacturer shall verify, in coordination with the qualifying activity, the capability of manufacturing products that meet the performance requirements of this specification. Refer to the qualifying activity for the guidelines necessary to retain qualification to this particular specification. The manufacturer shall immediately notify the qualifying activity at any time that the inspection data indicates failure of the qualified product to meet the performance requirements of this specification.

4.5.4.1 Retention of qualification for group A inspections. At intervals of not more than 12 months after initial qualification and every 12-month period thereafter, group A inspections for retention of qualification shall be performed and shall include the examinations and tests of table III. Sampling for group A inspection shall be as specified in 4.6.1.2. Manufacturers shall certify that there have been no material changes since initial qualification.

4.5.4.2 Retention of qualification for group B inspections. At intervals of not more than 24 months after initial qualification, and every 24-month period thereafter, group B inspections for retention of qualification shall be performed and shall include the examinations and tests of table IV. Sampling for group B inspection shall be as specified in 4.6.1.3. Manufacturers shall certify that there have been no material changes since initial qualification.

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4.5.4.3 Retention of qualification for group C inspections. At intervals of not more than 60 months after initial qualification and every 60-month period thereafter, group C inspections for retention of qualification shall be performed and shall include the examinations and tests of table V. Periodic inspection shall consist of group C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.4), delivery of products which have passed group A and group B inspections shall not be delayed pending the results of periodic inspection.

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TABLE II. Qualification inspection.

Inspection	Application		Requirement	Test method
	Cable splice	Fiber splice		
Group I (all sample units) [1/]				
Visual and mechanical				
Size	X	X	3.5.1	4.7.2.1
Weight	X	X	3.5.2	4.7.2.2
Identification marking	X	X	3.5.4	4.6.2.44.7.2.4
Workmanship	X	X	3.8	4.8
Screw threads	X	--	3.9	4.9
Optical				
Insertion loss	X	X	3.6.1	4.7.3.1
Return loss	X	X	3.6.3	4.7.3.3
Ambient light susceptibility	X	X	3.6.6	4.7.3.6
Crosstalk	X	--	3.6.5	4.7.3.5
Group II (two sample units)				
Cable pull out force	X	--	3.5.13	4.7.2.13
Fiber pull out force	--	X	3.5.12	4.7.2.12
Cable seal flexing (cable splice only)	X	--	3.5.5	4.7.2.5
Twist (cable splice only)	X	--	3.5.6	4.7.2.6
Impact	X	X	3.5.9	4.7.2.9
Crush	X	--	3.5.8	4.7.2.8
Axial compressive loading (cable splice only)	X	--	3.5.7	4.7.2.7
Vibration	X	X	3.5.10	4.7.2.10
Mechanical shock	X	X	3.5.11	4.7.2.11
Water pressure	X	--	3.7.6	4.7.4.5
Salt spray	X	--	3.7.5	4.7.4.4
Group III (two sample units)				
Thermal shock	X	X	3.7.3	4.7.4.2
Temperature/humidity cycling	X	X	3.7.4	4.7.4.3
Temperature cycling	X	X	3.7.12	4.7.4.11
Life aging	X	X	3.7.2	4.7.4.1
Insertion loss	X	X	3.6.1	4.7.3.1
Freezing water immersion	X	--	3.7.7	4.7.4.6
Fluid immersion [2/]	X	--	3.7.10	4.7.4.9
Sand and dust	X	--	3.7.8	4.7.4.7
Flammability	X	--	3.7.11	4.7.4.10

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TABLE II. Qualification inspection - Continued.

Inspection	Application		Requirement	Test method
	Cable splice	Fiber splice		
Group IV (lensed splices only - two sample units)				
Nuclear radiation resistance	--	X	3.7.9	4.7.4.8
Group V (polymeric parts only)				
Fungus resistance	X	X	3.3.9	4.7.2.14

NOTES:
1/ Visual and mechanical tests listed in this group shall be performed on unassembled splice parts. Assembled splices shall be used for the remaining groups.
2/ Each sample unit shall be tested in all fluids.

4.6 Conformance inspection. Conformance inspection shall include the examination of 4.7.2.1, 4.7.2.4, and 4.8, and the tests of 4.7.2.6 through 4.7.4.11, as applicable for the specific group.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.6.1.1 Unit of product. A unit of product shall consist of one splice (see 6.4.10).

4.6.1.1.1 Inspection lot. The inspection lot shall consist of the number of units of product that are offered for inspection at one time and of the same design as covered by one specification sheet (see 3.1). All of the units of product in the inspection lot submitted shall have been produced during the same production period with the same materials and processes.

4.6.1.1.2 Sample unit. A sample unit shall consist of a unit of product selected at random from the inspection lot.

4.6.1.1.3 Sample unit preparation. No preparation of the sample unit is required for group A inspections. Unless otherwise specified herein, sample unit splices for groups B and C inspections shall be fully assembled utilizing two lengths of cable, each at least 5 meters long, compatible with the splice undergoing testing. The cable shall contain at least two fibers for those units that have the capability to splice more than one fiber (i.e., multisplice units). At least two adjacent splices shall be made per unit. When optical crosstalk tests are required, crosstalk shall be measured between fiber splices enclosed in the same unit. The non-spliced fiber ends of the sample unit shall be optically finished to permit optical assessment of the splice.

4.6.1.1.4 Specimen. A specimen shall be a sample unit that has been prepared in accordance with 4.6.1.1.3.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspection tests specified in table III. The inspection tests may be performed in any convenient order. Refer to 4.5.4.1 for the time intervals of group A inspections for retention of qualification.

4.6.1.2.1 Sampling plan. Tests shall be performed on 100 percent of the product supplied under this specification.

4.6.1.2.2 Failures. One or more failures shall constitute group A inspection failure of the sample unit.

4.6.1.2.3 Disposition of sample units. Sample units that have failed any of the group A inspection tests shall not be shipped or submitted for group B testing.

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TABLE III. Group A conformance inspection.

Inspection	Application		Requirement	Test method
	Cable splice	Fiber splice		
Visual and mechanical				
Size [1/]	X	X	3.5.1	4.7.2.1
Identification marking [1/]	X	X	3.5.4	4.7.2.4
Workmanship	X	X	3.8	4.8
NOTE: [1/] In-process controls may be used by the manufacturer in lieu of this group A test.				

4.6.1.3 Group B inspection. Group B inspection shall consist of the optical test specified in table IV and shall be performed on sample units that have been subjected to and have passed the group A inspection. Refer to 4.5.4.2 for the time intervals of group B inspections for retention of qualification.

TABLE IV. Group B conformance inspection.

Inspection	Requirement	Test method
Insertion loss	3.6.1	4.7.3.1

4.6.1.3.1 Disposition of samples. Samples subjected to group B inspection shall not be shipped.

4.6.1.3.2 Sampling plan. Six specimens shall be selected from a lot with the same PIN within 24 months after the notification date of qualification and during every 24-month period thereafter, except when the total production in a 24-month period is less than 250 units of product or a total of 60 months have elapsed since the inspection was performed, in which case only three specimens shall be tested.

4.6.1.3.3 Failures. If one or more sample units fail to pass group B inspection, the lot from which the samples were selected shall be rejected.

4.6.1.3.4 Rejected lots. Requirements regarding the rework of rejected lots shall be as identified in the qualification instructions (see 6.7.1).

4.6.2 Group C inspection. Group C inspections shall consist of the inspection tests as specified in table V. The inspections shall be performed in the order shown on sample units that have passed group A and B inspections. Refer to 4.5.4.3 for the time intervals of group C inspections for retention of qualification.

4.6.2.1 Disposition of sample units. Sample units that have been subjected to group C inspection shall not be shipped.

4.6.2.2 Sampling plan. Every 60 months, eight sample units with the same PIN that have passed group B inspection shall be selected.

4.6.2.3 Failures. Failure of one or more specimens or sample units shall constitute group C inspection failure.

4.6.2.4 Noncompliance. Requirements regarding failure of group C inspection shall be as identified in the qualification instructions (see 6.7.2).

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TABLE V. Group C conformance inspection.

Inspection	Application		Requirement	Test method
	Cable splice	Fiber splice		
Group I (all sample units)				
Optical				
Insertion loss	X	X	3.6.1	4.7.3.1
Return loss	X	X	3.6.3	4.7.3.3
Crosstalk	X	--	3.6.5	4.7.3.5
Group II (four sample units)				
Cable pull out force	X	--	3.5.13	4.7.2.13
Fiber pull out force	--	X	3.5.12	4.7.2.12
Axial compressive loading (cable splice only)	X	--	3.5.7	4.7.2.7
Twist (cable splice only)	X	--	3.5.6	4.7.2.6
Impact	X	X	3.5.9	4.7.2.9
Crush	X	--	3.5.8	4.7.2.8
Mechanical shock	X	X	3.5.11	4.7.2.11
Group III (four sample units)				
Temperature/humidity cycling	X	X	3.7.4	4.7.4.3
Temperature cycling	X	X	3.7.12	4.7.4.11
Life aging	X	X	3.7.2	4.7.4.1
Insertion loss	X	X	3.6.1	4.7.3.1

4.7 Methods of inspection.

4.7.1 Equivalent test methods. The use of equivalent test methods is allowed provided the preparing activity and the qualifying activity have approved the use of that equivalent test method by that manufacturer (see 6.3.4).

4.7.2 Visual and mechanical examination. The splice and splice parts shall be examined in accordance with TIA-455-13 and with the data recording and evaluation criteria in Measurements 3101 through 3105 of MIL-STD-1678-3 to verify that materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.3, 3.4, and 3.5).

4.7.2.1 Size. Each dimension identified in the specification sheet (see 3.1) for the splice parts shall be measured using calibrated measuring devices with the range, precision, and accuracy appropriate for the tolerances specified (see 3.5.1) and conform to the setup and measurement process (test procedure) and applicable data sheet content in accordance with Measurement 3101 of MIL-STD-1678-3.

4.7.2.2 Weight. The splice or splice parts shall be weighed using calibrated scales having the range, precision, and accuracy appropriate for the tolerances specified (see 3.5.2) and conform to the setup and measurement process (test procedure) and applicable data sheet content in accordance with Measurement 3102 of MIL-STD-1678-3.

4.7.2.3 Color. The color of the splice or splice parts shall be visually inspected for compliance with 3.5.3 and conform to the setup and measurement process (test procedure) and applicable data sheet content in accordance with Measurement 3104 of MIL-STD-1678-3.

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4.7.2.4 Identification markings. Identification markings on the splice parts shall be visually examined and measured for conformance to 3.5.4 and conform to the setup and measurement process (test procedure) and applicable data sheet content in accordance with Measurement 3104 of MIL-STD-1678-3. Inspection for legible markings shall be performed from a distance of at least 6 inches with normal room lighting and without the aid of magnification.

4.7.2.5 Cable seal flexing (cable splice only). The splice shall be tested in accordance with TIA/EIA-455-1. After test exposure, the assemblies shall be visually examined for seal damage. The cable splice assembly shall be exposed to 100 flexing cycles, the assembly rotated approximately 90 degrees in the flexing fixture, and then exposed to another 100 flexing cycles (see 3.5.5). The cycling rate is not to exceed 14 cycles per minute if the test is performed manually.

4.7.2.6 Twist (cable splice only). The splice shall be tested in accordance with TIA-455-36 for 50 cycles. One cycle shall consist of a 360-degree twist ±180 degrees about the neutral axis. The tensile load shall be 10 N (2.2 lbs) for single fiber cable and 50 N (11 lbs) for multiple fiber cable and the number of loads to be applied shall be one (see 3.5.6). The change in optical transmittance shall be measured during and after the test (see 4.7.3.2). Change in optical transmittance measurements shall be made after the 10th, 25th, and 50th cycle. At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.7 Axial compressive loading (cable splice only). Axial compressive loading shall be tested as specified herein. If the two ends of the splice assembly are essentially identical, only one end of each splice assembly needs to be tested (see 3.5.7).

a. The test cable forming a part of the assembled splice shall be gripped over a length equal to at least three cable diameters starting one cable diameter from the splice enclosure or the strain relief if it is external to the enclosure. The splice enclosure shall be gripped so as not to alter any of its mechanical properties.

b. The gripped cable and splice enclosure shall be forced together along the direction of the cable axis where the cable enters the splice assembly. The force magnitude is a function of the cable diameter as listed in table VI. Loading and unloading rates are such that the splice is compressively loaded and unloaded linearly in time with the loading and unloading times each less than 10 seconds. The time spent under full compressive load shall be at least 1 minute. No tensile load shall be applied.

c. Any suitable test fixture may be used provided it properly grips the splice enclosure and cable, allows control and measurement of the applied compressive axial force, and allows visual observation of the region where the cable meets the splice assembly. The test fixture shall also allow for the performance of any optical tests specified herein while the splice assembly is subject to the compressive axial load.

d. With load applied, measure the distance between the fixture clamp and the enclosure or strain relief and measure the insertion loss.

e. Remove the load.

f. Visual observations and examinations shall be made with 3X magnification.

TABLE VI. Force magnitude.

Nominal diameter (millimeters)	Compressive axial force (Newton)
0 to 2.9	10
3 to 5.9	20
6 to 9.9	50
10 to 19.9	100
20 and larger	200

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4.7.2.8 Crush. Cable splices shall be tested in accordance with TIA-455-26 with the exception that the test sample shall be a splice (see 3.5.8). The test load shall be 1,250 N and the number of loading cycles shall be one. The two load bearing surfaces may be faced with 1-inch thick rubber pads with a durometer between 60 and 75. The cable splice shall be positioned on the crush pads so that at least half the cable splice is on the crush pads. The middle of the cable splice shall be placed at the center of the crush pads. The change in optical transmittance shall be measured during and after the test (see 4.7.3.2). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.9 Impact.

4.7.2.9.1 Cable splice. Cable splices shall be tested in accordance with Method A of TIA-455-2, moderate service class (see 3.5.9). The change in optical transmittance shall be measured after the test in accordance with 4.7.3.2 (see 3.5.9). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.9.2 Fiber splice. Fiber splices shall be tested in accordance with Method C of TIA-455-2. The change in optical transmittance shall be measured after the test in accordance with 4.7.3.2 (see 3.5.9). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.9.3 Mechanical splice. Mechanical splices shall be tested using a fixture as shown on figure 1 that contains a steel block with a 1.2-centimeter (cm) (0.5-inch) minimum thickness. Mechanical splice cabling (both cable ends) shall be secured to a clamp 1.5 meters (m) (59 inches) from the back of the mechanical splice strain relief. With the mechanical splice hanging under its own weight, the mechanical splice shall be at the horizontal and vertical center of the steel block (see figure 1). The mechanical splice shall be raised 1.5 m (59 inches) to the height of the clamp and released with the cable extended so as to strike the steel block. The test shall consist of eight drops. The change in optical transmittance shall be measured after the test in accordance with 4.7.3.2 (see 3.5.9). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.10 Vibration. Splices shall be tested in accordance with test Condition B, Measurement 3201.1 of MIL-STD-1678-3. Sub-test Conditions shall be performed as specified (see 3.1). Where not specified, sub-test Conditions II and VI shall be performed. Optical discontinuity shall be monitored during and after the test in accordance with 4.7.3.4 (see 3.5.10). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.11 Mechanical shock. Splices shall be tested in accordance with grade A, class I, type A of MIL-DTL-901 and Measurement 3202.1 of MIL-STD-1678-3. Optical discontinuity shall be monitored during the test (see 4.7.3.4), and the change in optical transmittance shall be monitored after the test in accordance with 4.7.3.2 (see 3.5.11). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.12 Fiber pull out force. The fiber pullout force shall be tested by applying the axial tensile load specified between the fiber and the fiber splice housing for a duration of 1 minute. The change in optical transmittance shall be monitored during and after the test in accordance with 4.7.3.2 (see 3.5.12). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.13 Cable pull out force. Cable splices shall be tested in accordance with TIA-455-6. The axial tensile load shall be applied up to the load specified and shall be maintained for 10 minutes for cable-to-cable splice closure or 1 minute for splices attaching directly to fiber cable components. The change in optical transmittance shall be measured during and after the test in accordance with 4.7.3.2 (see 3.5.13). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.2.14 Fungus resistance. Splices composed of materials not listed as fungus inert in Guideline 4 of MIL-HDBK-454 shall be tested in accordance with Measurement 3401 of MIL-STD-1678-3 for a duration of 28 days (see 3.3.9).

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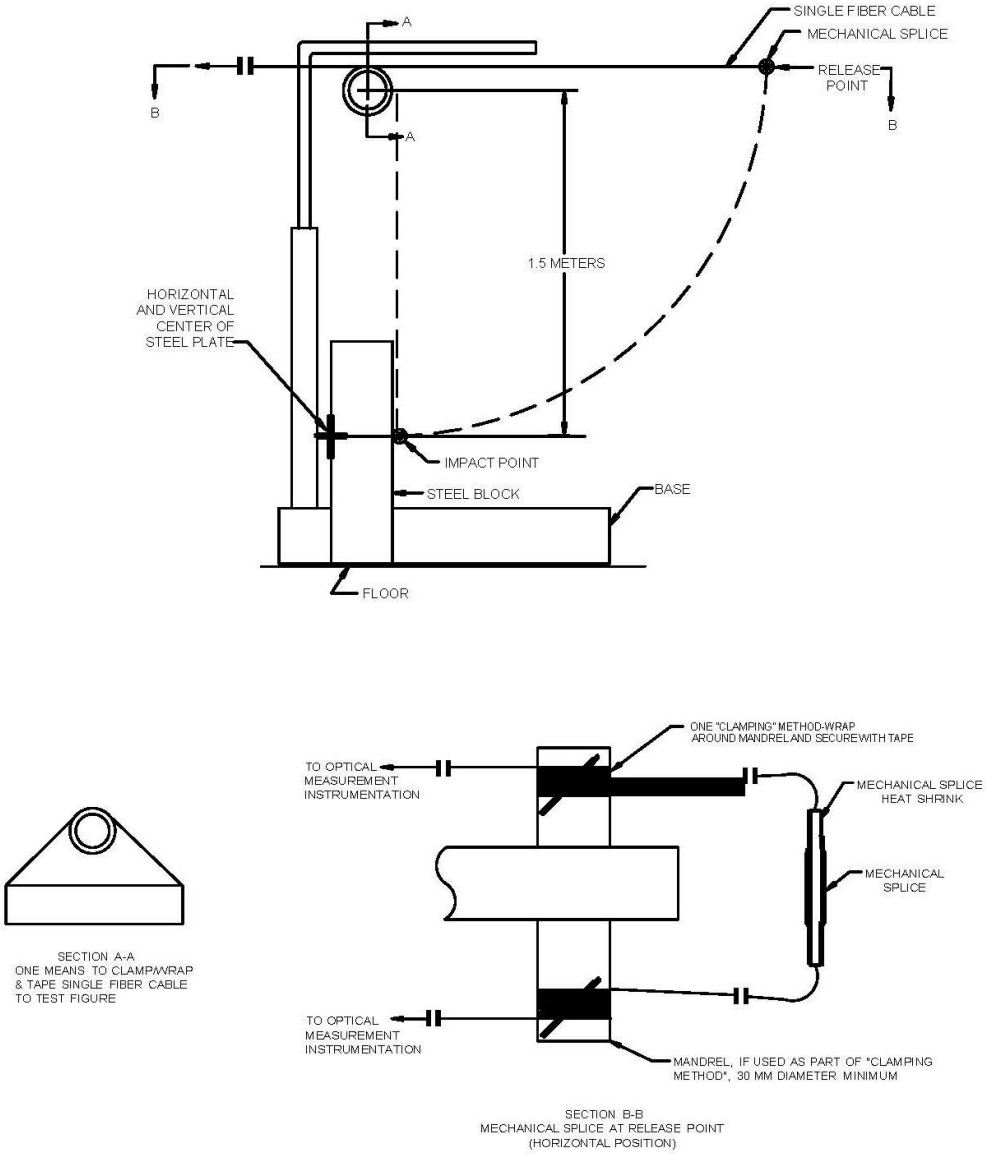


FIGURE 1. Mechanical splice impact fixture and test setup.

4.7.3 Optical conformance test methods. Optical measurements performed in accordance with an applicable TIA-455 series standard shall adhere strictly to the setup and test procedure specified in the applicable measurement in Measurements 2101 through 2106 of MIL-STD-1678-2. Multiple fibers may not be concatenated during the measurement of change in optical transmittance or optical discontinuity. In splices with one, two, or three optical channels, optical measurements shall be made simultaneously on each channel (see 3.6). Unless otherwise specified herein, in splices with four or more channels, optical measurements shall be made simultaneously on three randomly selected channels. The center wavelength and light launch conditions for specified fiber size shall be in accordance with MIL-STD-1678-2 and as specified in table VII.

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TABLE VII. Light launch conditions.

Fiber type	Launch condition
Single mode	30-mm diameter mandrel
Multimode	Uniform overfill (initial insertion loss only) and 70/70 restricted or equivalent

4.7.3.1 Insertion loss. The initial insertion loss of multimode splices shall be measured in accordance with Measurement 2101 of MIL-STD-1678-2 using both an overfill launch condition and a restricted launch that is in accordance with Measurement Support Process 2203 of MIL-STD-1678-2. Subsequent insertion loss tests shall be performed with the specified restricted launch condition in accordance with Measurement Support Process 2203 of MIL-STD-1678-2. The insertion loss of single mode splices shall be measured in accordance with Measurement 2101 of MIL-STD-1678-2 using light launch conditions that are in accordance with Measurement Support Process 2203 of MIL-STD-1678-2.

4.7.3.2 Change in optical transmittance. The change in optical transmittance shall be measured in accordance with Measurement 2102 of MIL-STD-1678-2 or by an equivalent method. A monitor fiber shall be used to evaluate the change in transmittance due to exposure of the splice to environmental and physical tests (see 3.6.2). The periodicity of the measurements shall be sufficient for the environmental test performed as approved by the qualifying activity. The use of a reference fiber to evaluate the change in optical transmittance due to exposure of the cable to the environmental tests is optional.

4.7.3.3 Return loss. The return loss (see 6.4.9) shall be measured in accordance with Measurement 2105 of MIL-STD-1678-2 or by an equivalent method (see 3.6.3).

4.7.3.4 Optical discontinuity. Splices shall be tested in accordance with Measurement 2104 of MIL-STD-1678-2 (see 3.6.4) using equipment having a time resolution sufficient to resolve discontinuities (see 6.4.5) of duration not less than 50 microseconds (µs). For tests of extended duration, discontinuity measurements may be made at discrete times during the test as approved by the qualifying activity.

4.7.3.5 Crosstalk. Crosstalk (see 6.4.4) shall be measured in accordance with Measurement 2103 of MIL-STD-1678-2 (see 3.6.5).

4.7.3.6 Ambient light susceptibility. Ambient light susceptibility (see 6.4.1) shall be tested in accordance with TIA-455-22 (see 3.6.6) using the following exceptions: The measurement shall be performed at the maximum specified irradiance of 100 mW/cm^2, the power measured in the “on” state shall be referenced to 1 mW, and the test temperature shall be 25±5 °C.

4.7.4 Environmental test methods. The environmental tests in table II shall be performed in the sequence listed. The environmental tests shall be performed in accordance with the applicable TIA-455 series or otherwise cited test standards with the exception of adhering strictly to any deviations, constraints, or augmentations in the applicable 3300 series Measurements in MIL-STD-1678-3. Post-exposure optical transmittance measurements may be taken up to 24 hours after completion of the environmental exposure.

4.7.4.1 Life aging. Splices shall be tested in accordance with Measurement 3303 of MIL-STD-1678-3 and as specified herein (see 3.7.2). The following special test conditions and modifications to Measurement 3303 of MIL-STD-1678-3 shall apply to these tests:

a. The specimen shall be exposed to dry air at 85 °C, -0 °C/+3 °C for a period of 240 hours. For space application, the test duration shall be 1,000 hours.

b. Only the splice and the portions of the test cables nearest the splice shall be exposed to the test environment.

c. Pre-test and post-test measurements of the splice outer diameter shall be taken and reported.

d. Visual inspection of the splice shall be conducted using 3X magnification.

e. Monitor change in optical transmittance (see 4.7.3.2).

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4.7.4.2 Thermal shock. Splices shall be tested in accordance with Measurement 3304 of MIL-STD-1678-3 (see 3.7.3). The temperature extremes shall be the specified non-operational temperature extremes (see 3.1). The change in optical transmittance shall be measured after the test in accordance with 4.7.3.2. At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.4.3 Temperature/humidity cycling. The splices shall be tested in accordance with Measurement 3302 of MIL-STD-1678-3. The sub-cycle shall be included in the test. The change in optical transmittance shall be monitored during and after the test in accordance with 4.7.3.2 (see 3.7.4). At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.4.4 Salt spray (corrosion) (cable splices only). The splices shall be tested in accordance with Measurement 3402.1 of MIL-STD-1678-3 for 500 hours (see 3.7.5). The exposure temperature shall be 35 °C. After test exposure, the splices shall be externally cleaned and examined for possible salt penetration into the splice watertight (see 6.4.15) areas (the enclosure shall be removed).

4.7.4.5 Water pressure (cable splices only). The splices shall be tested for water pressure susceptibility by immersion in fresh water to a depth of 10.4 meters for a period of 48 hours (see 3.7.6). The water temperature shall be maintained between 10 and 35 °C during the exposure period. The change in optical transmittance shall be monitored after the test in accordance with 4.7.3.2. The splices shall be externally cleaned and examined (with enclosures removed) for possible water penetration into the splice watertight areas.

4.7.4.6 Freezing water immersion (ice crush). Cable splices shall be tested in accordance Measurement 3307 of MIL-STD-1678-3 (see 3.7.7). The size of the water vessel shall be such that when the cable splice is placed in the center of the vessel, the cable splice is within 150 mm of the sides, top, and bottom. The change in optical transmittance shall be monitored during and after the test in accordance with 4.7.3.2. At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.7.4.7 Sand and dust. Cable splices shall be tested in accordance with TIA-455-35 (see 3.7.8) except as noted herein. Each cable splice shall be oriented in the chamber such that the longitudinal (long) axis is in line with the oncoming air flow. The cable splice shall be affixed in such a manner that the orientation of the cable splice does not change throughout the duration of the test. The 16-hour holding period of step 5.5 is not required. Step 5.6 may proceed immediately after reaching temperature stabilization. The change in optical transmittance shall be measured before the dust test, during the 6-hour exposure period of step 5.4, before step 5.6, during the 6-hour exposure period of step 5.6, and after the test. At the completion of the test and after cleaning, the splice shall be visually examined in accordance with 4.7.2.

4.7.4.8 Nuclear radiation resistance. Lensed splices shall be tested for susceptibility to gamma radiation in accordance with TIA-455-64 (see 3.7.9). The splice shall be subjected to a total dose as specified by the qualifying activity at a dose rate of 50+0, -20 rads (Si)/sec. The change in optical transmittance (see 4.7.3.2) shall be measured during and after the test. The test shall be conducted at the low operating temperature and at 25±5 °C. If the change in optical transmittance at 25±5 °C is greater than the change in optical transmittance at the low operating temperature, the splice shall be tested at the high operating temperature.

4.7.4.9 Fluid immersion. Cable splices shall be tested in accordance with Measurement 3409.1 of MIL-STD-1678-3, temperature range 1, with the following exceptions: exposure to automobile gasoline is not required and the test temperature for lubricating oil exposure shall be 73 to 77 °C (see 3.7.10). Sample preconditioning shall be done under ambient conditions for at least 4 hours. After the 24-hour immersion in each fluid, the splice shall be cleaned and examined for possible fluid penetration. The splice and sample of each polymeric material shall be completely dried after each immersion and after performing the examination but prior to the next immersion.

4.7.4.10 Flammability. Cable splices shall be tested in accordance with EIA-364-81 except that the flame shall be applied for 60 seconds during the second flame application and only one sample shall be tested (see 3.7.11). A flame with a 38.1-millimeter (1.50-inch) flame height shall be applied to the approximate center of the cable splice. The cable splice shall be configured with a full complement of fiber splices. No conditioning is required before the test. The change in optical transmittance shall be measured during and after the first application of the flame in accordance with 4.7.3.2.

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4.7.4.11 Temperature cycling. Splices shall be tested in accordance with Measurement 3301 of MIL-STD-1678-3 using the test condition schedule and soak times in accordance with Measurement 3301 (see 4.7.2.12). The change in optical transmittance shall be measured during and after the test in accordance with 4.7.3.2. At the completion of the test, the splice shall be visually examined in accordance with 4.7.2.

4.8 Workmanship. Splice parts shall be visually examined in accordance with TIA-455-13 to verify that they meet the workmanship requirements of 3.8, conform to the setup and measurement process (test procedure), and include the applicable data sheet content in accordance with Measurement 3103 of MIL-STD-1678-3.

4.9 Screw threads. Screw threads shall be checked after protective coating by means of ring and plug gauges (see MIL-STD-1373 for guidance) to verify that it meets screw thread requirements of 3.9.

4.10 Toxicity. When specified (see 3.1), a Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.1 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the materials using data provided by the manufacturer/distributor to the NMCPHC (see 3.3.1 and 6.5).

4.11 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3.2 as required by the qualifying activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The fiber optic cable splices covered by this specification are intended for use in military applications where their performance characteristics are required. The splices covered by this specification are unique due to the fact that these devices must be able to operate satisfactorily in systems under the following demanding conditions: 10 g's vibration and over 1,000 g's shock. In addition, these requirements are verified under a qualification system. Commercial splices are not designed to withstand these environmental conditions. Figure 2 is an example of a generic splice assembly with splice parts identified.

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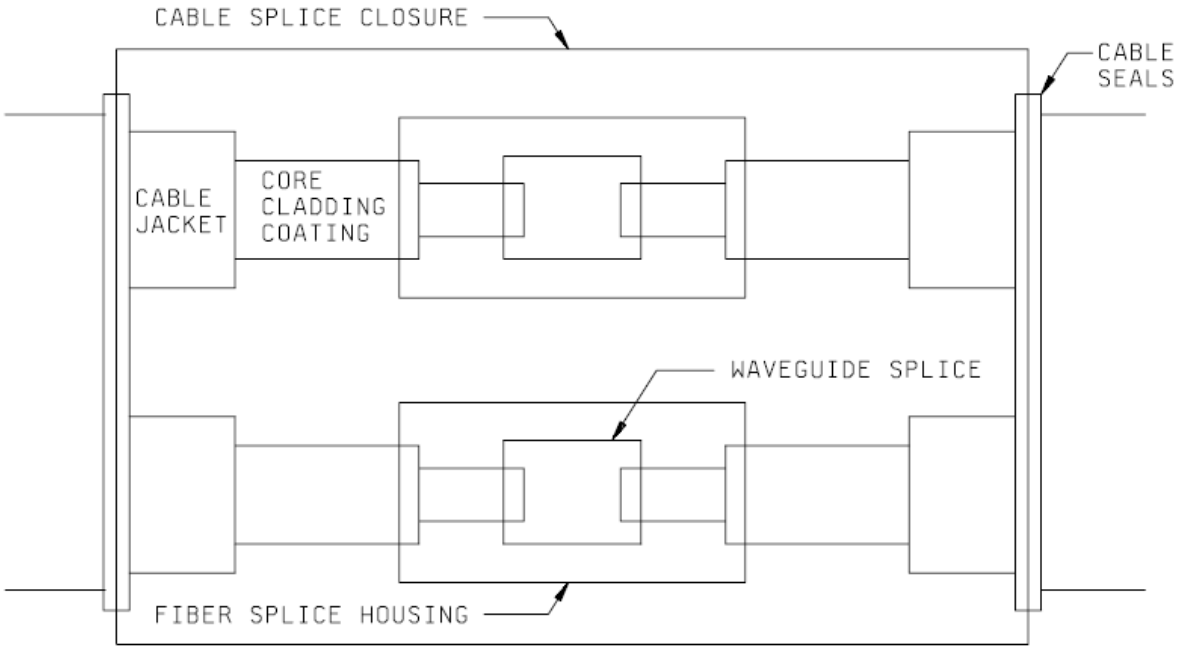


FIGURE 2. Typical splice assembly.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the specification sheet.

c. Quantity of splices required.

d. Inclusion of splicing tools, if desired (see 3.4.8).

e. Packaging requirements (see 5.1).

f. Exception, if any, to the optional provisions of this specification, including responsibility for inspection.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24623 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160, CommandStandards@navy.mil; however, information pertaining to qualification of products may be obtained from the agent for the qualifying activity: DLA Land and Maritime; VQP, P.O. Box 3990, Columbus, OH 43218-3990. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Copies of SD-6 “Provisions Governing Qualification” are available online at https://quicksearch.dla.mil.

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6.3.2 Verification program. A verification program should be established and maintained in accordance with MIL-STD-790 or equivalent standard. Evidence of such compliance will be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. The verification system procedures, planning, and all other documentation and data that comprise the verification system should be available to the Government for review. The Government may perform any necessary inspections, verifications, and evaluations to ascertain conformance to the requirements and the adequacy of the implementing procedures (see 4.).

6.3.3 Test equipment and inspection facilities. Provision for test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspections should be the responsibility of the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment should be in accordance with ISO/IEC 17025 or equivalent standard (see 4.3).

6.3.4 Alternate forms of conformance inspection and equivalent test methods. Requests for an alternate form of conformance inspection should be submitted to the qualifying activity and to the preparing activity (see 4.66). Alternate forms of conformance inspection may be used upon written approval by the qualifying activity and by the preparing activity. The use of equivalent test methods is allowed (see 4.7.1). The manufacturer should have conducted both test methods and have submitted complete test data to the preparing activity and to the qualifying activity verifying the equivalency of each equivalent test method proposed.

6.4 Definitions. Definitions of terms are in accordance with TIA-440 and as stated below.

6.4.1 Ambient light susceptibility. Ambient light susceptibility is the interference of the optical data signals in a splice by ambient optical power incident upon its exterior. The purpose of this optical performance test is to confirm the ability of a splice to exclude ambient optical power from its optical signal circuits.

6.4.2 Cable splice. A cable splice is that portion of the splice that holds together the cable jackets and strength members for purposes of protecting the fibers from physical stresses.

6.4.3 Closure. The closure is the portion of a cable splice that covers the fiber splice housing, seals against the outer jackets of the joined cables, provides protection against the environment, and provides mechanical strength for the joint.

6.4.4 Crosstalk. Crosstalk is unwanted, coupled optical energy from an optical signal circuit (the active channel) into another optical signal circuit or group of signal circuits (the passive channels). The purpose of this optical performance test is to ensure that the splice circuits have adequate optical channel isolation.

6.4.5 Discontinuity. Discontinuity refers to measuring the presence of relatively sharp, short duration interruptions or dropouts of the optical signals transmitted by the splice. The purpose of this optical performance test is to confirm the splice's signal discontinuities are not excessive when subjected to physical stress tests.

6.4.6 Fiber splice. A fiber splice is that portion of a splice that joins together two fibers and provides physical protection for them.

6.4.7 Housing. A housing, such as tapes, jackets, coatings, and other components necessary for attaching, supporting, and aligning fibers, applied over a fiber optic splice and proximate fiber for their protection against the environment, for some mechanical strength, and for preservation of the integrity of the fiber coating, buffer, cladding, and core. The housing includes a means for mounting the aligned assembly in an interconnection box or cable splice closure.

6.4.8 Insertion loss. Insertion loss is the total optical signal power loss in an optical circuit caused by inserting a splice into the optical circuit. The purpose of this optical performance test is to confirm that a splice does not excessively attenuate the optical signal.

6.4.9 Return loss. Return loss is the optical power (dB) that is reflected back towards the source of optical power by the splice.

6.4.10 Splice. A splice is a generic term for a device that permanently joins optical fibers and cables in a protective manner and may consist of several parts.

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6.4.11 Splice enclosure. A splice enclosure is the outermost protective structure of a splice.

6.4.12 Splice junction. A splice junction is the bonded optical interface of mating fibers.

6.4.13 Splice kit. A splice kit consists of all splice parts that are needed to produce a splice.

6.4.14 Splice parts. Splice parts are the individual splicing elements (see 6.4.12) of the splicing kit (see 6.4.13).

6.4.15 Watertight. The watertight property of a splice is its ability to prevent leakage of water into the internal splice parts.

6.4.16 Waveguide splice. A waveguide splice is a permanent joint between the transmission elements of two waveguides so that signals may pass from one waveguide to the other with minimal loss, for example, a joint between the cores and claddings of two optical fibers.

6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.6 Safety note. Care must be taken when handling the very fine (small diameter) optical fibers to prevent skin puncture or contact of the fiber with the eye. Also, direct viewing of the optical terminal face of a terminated cable while it is propagating optical energy is not recommended unless prior assurance has been obtained as to the safe power output of the terminal.

6.7 Conformance inspection.

6.7.1 Rejected lots. If a group B inspection lot is rejected, the contractor may rework it to correct the defects or screen out the defective units (if applicable) and resubmit the lot for inspection.

6.7.2 Noncompliance. If a sample fails to pass group C inspection, the manufacturer should notify the qualifying activity of the failure and take corrective action on the materials or processes or both, as warranted, and on all units of product that can be corrected, that were manufactured under essentially the same conditions with essentially the same materials and processes, and that are considered subject to the same failure. Acceptance of the product should be discontinued until corrective action that is acceptable to the Government has been taken. After the corrective action has been taken, group C inspection should be repeated on additional sample units (all inspections tests or the inspection test that the original sample failed at the option of the Government). Group A and B inspections may be reinstituted; however, final acceptance should be withheld until the group C re-inspection has shown that the corrective action was successful. In the event of failure after re-inspection, information concerning the failure and corrective action taken should be furnished to the cognizant inspection activity and the qualifying activity.

6.8 Subject term (key word) listing.

Closure

Housing

Joint

Return loss

Splice, fusion

Splice, mechanical

Watertight

Waveguide

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – CR
Navy – SH
Air Force – 85
NASA – NA

Review activities:
Army – AR, MI
Navy – AS, CG, EC, MC
Air Force – 19, 70, 99
DLA – CC

Preparing activity:
Navy – SH
(Project 6060-2016-006)

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